

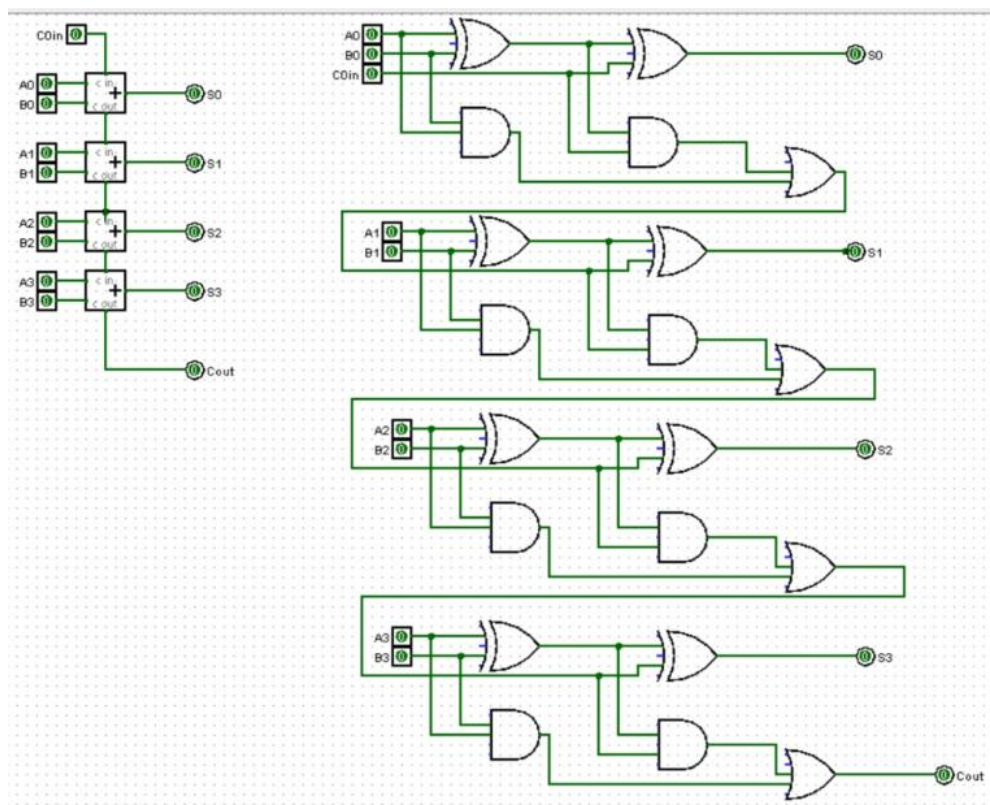
Lab 3: Implementation of 4-bit Ripple Carry Adder

Aim

To design and simulate a 4-bit Ripple Carry Adder using Logisim Evolution and verify its operation.

Procedure

1. Open Logisim Evolution.
2. Design a Full Adder circuit or use a built-in Full Adder component.
3. Create four Full Adder blocks for 4-bit addition.
4. Provide two sets of 4-bit inputs (A0–A3 and B0–B3).
5. Connect the carry-out of each Full Adder to the carry-in of the next stage.
6. Set the initial carry-in (C_{in}) of the first Full Adder to 0.
7. Connect outputs (S0–S3) and final carry-out (C_{out}) to LEDs.
8. Apply different binary input combinations and observe the output.



Result

The 4-bit Ripple Carry Adder was successfully designed and simulated using Logisim Evolution.

- The circuit correctly performed binary addition of two 4-bit numbers.
- Proper carry propagation was observed from one stage to the next.
- The outputs matched the expected theoretical results for all tested inputs.